Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. All experiments were run on a single GPU with 16 GB of memory. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the accuracy of Transformer by dataset.

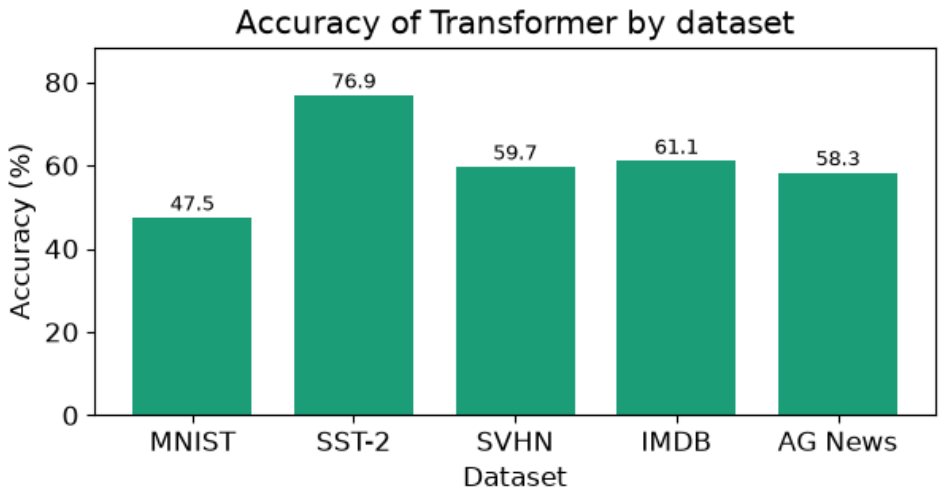


Figure 1: Accuracy of Transformer by dataset for the models in this experiment.

On IMDB, Transformer obtains a accuracy of 69.9% (Figure 1). Transformer posts over 55% accuracy when tested on AG News.

Figure 2 shows the training loss.

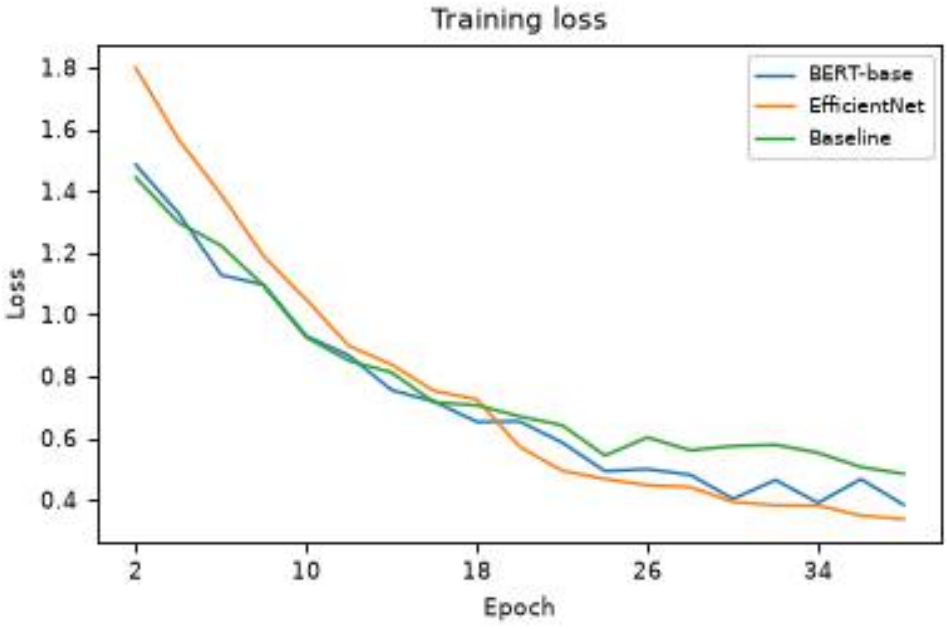


Figure 2: Training loss for the models in this experiment.

Epoch 12 sees BERT-base post a loss of 0.867. As shown in Figure 2, BERT-base has a loss of 1.096 at epoch 8. As shown in Figure 2, By epoch 12, the loss of EfficientNet stands at 0.577. Epoch 30 sees BERT-base post a loss of 0.435.

Figure 3 shows the validation accuracy by dataset.

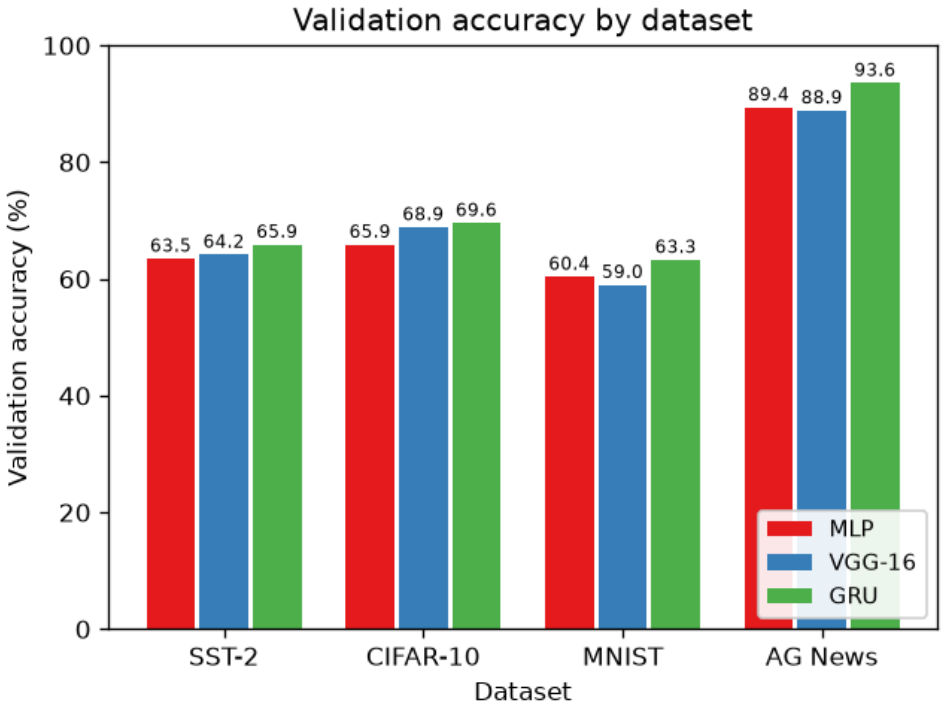


Figure 3: Validation accuracy by dataset for the models in this experiment.

Evaluated on SST-2, VGG-16 comes in at about 60% validation accuracy. GRU posts 98.6% validation accuracy when tested on AG News. On SST-2, VGG-16 obtains a validation accuracy of 64.2%. As shown in Figure 3, MLP posts 65.4% validation accuracy when tested on MNIST.

Figure 4 shows the training loss.

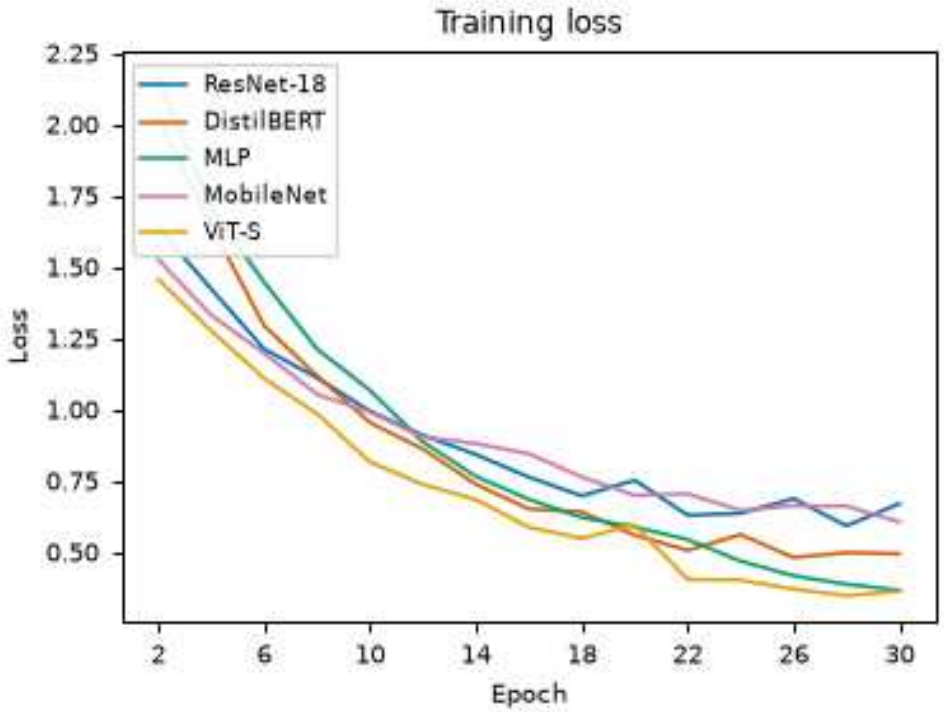


Figure 4: Training loss for the models in this experiment.

By epoch 18, the loss of MobileNet stands at over 0.7. As shown in Figure 4, Epoch 8 sees MobileNet post a loss of 1.05. DistilBERT has a loss of 1.040 at epoch 18.

3 Discussion

On MNIST, XGBoost obtains an accuracy of 53.4%. The differences between the models are discussed in

the next section.